In past technological revolutions, most economic value flowed to those who used the technology, not to those who built the infrastructure, writes Umar Ruhi. *Photograph courtesy of Amrulqays Maarof, Pixabay.com*

Canada's AI challenge is readiness, not infrastructure

Governance capacity and workforce preparedness, not compute spending, will determine whether Canada captures AI's economic promise.

Umar Ruhi

Opinion

Canada is caught in a high-stakes paradox. We are investing billions of dollars in the "pipes" of the 21st-century economy—sovereign compute, hyperscale data centres, and research clusters—while the "machinery" of our businesses remains stalled. Business AI adoption has reached only 12 per cent, and although more than half of Canadian employees are experimenting with generative AI, only two per cent of business leaders report a measurable return on those investments.

We are, in effect, building a multi-billion-dollar superhighway for a nation that hasn't yet learned to drive.

The federal government has committed hundreds of millions of dollars to sovereign compute infrastructure. Microsoft is investing $19-billion in AI infrastructure through 2027, and Alberta has announced a $100-billion AI data centre strategy. While these investments are necessary, compute capacity is rapidly becoming a commodity. What will differentiate economies is institutional readiness: organizational capacity, a skilled workforce, and governance frameworks that translate AI investment into broad-based economic value.

OECD's 2025 economic survey notes that this country's innovation policy focuses too heavily on research inputs rather than on adoption and firm-level diffusion. In the case of AI, the leading barriers to adoption are a lack of knowledge and skills. Canada ranks 44th out of 47 countries in AI literacy and training. Canadian workers want to learn, but their employers and institutions have been slow to equip them. Statistics Canada data show that while AI use by domestic businesses doubled over the past year, growth is slowing. In several sectors, firms that tried AI are pulling back. Among those that persist, priorities include developing new workflows and training existing staff, i.e., focusing on institutional readiness activities.

In past technological revolutions, most economic value flowed to those who used the technology, not to those who built the infrastructure. Electricity, invented in the 1880s, did not transform manufacturing until factories were redesigned around it decades later. The gains from the internet went to firms that reimagined their operations and created new business models. Following the same pattern, competitive advantage from AI will belong to countries whose workers, managers, and institutions are ready to experiment, adopt, and integrate. Canada cannot afford to build first and adopt later. Both must happen together.

Institutional readiness also reframes regulation and innovation. When governance is understood as a condition for productive adoption rather than an obstacle, the two can move together. The Treasury Board's directive on automated decision-making offers a working example. It requires impact assessments, transparency, and recourse for AI systems that affect people's rights, while keeping research and experimentation out of scope, so the rules do not burden exploration. That scaffolding, however, applies only to the federal public service. Fifteen months after the AI and Data Act died with Bill C-27, no successor legislation has been tabled. Canada remains the only G7 country without a comprehensive AI law in force or under active legislative consideration, and this uncertainty stalls organizations seeking clear rules before committing to adoption.

The workforce picture is more nuanced than headlines suggest. While 60 per cent of workers are in occupations highly exposed to AI-related transformation, most businesses that have adopted AI report no change in employment. Furthermore, roughly half of exposed workers are in roles where AI is more likely to reshape their tasks than to eliminate their positions. Canada's immediate challenge is job redesign, managerial adaptation, and practical training for workers whose roles will shift before entire occupations disappear.

Three concrete policy measures would help.

First, governments should tie support for AI adoption, procurement decisions, and public funding to readiness requirements, asking organizations to demonstrate plans for risk assessment, privacy protection, AI disclosure, and worker transitions.

Second, Canada should launch a national AI readiness initiative focused on mid-career re-skilling, manager education, and sector-specific transition planning.

Third, public policy should encourage organizations to move beyond using AI for routine task automation and towards redesigning business processes around AI capabilities where the larger productivity gains lie. Incentives tied to process innovation—not just technology acquisition—would accelerate this shift.

Canada was the first country to adopt a national AI strategy in 2017. That early advantage has eroded. The path forward requires treating governance capacity and workforce readiness as upstream determinants of whether AI investment translates into economic value, and investing in them accordingly.

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